# NASIR MAHMOOD ABBASI, PhD

Bioinformatician

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## RESEARCH PROFILE

Bioinformatician with a PhD in Cancer Biology, working at the interface of computational analysis and experimental biology. Doctoral work applied single-cell multi-omics to characterize tumor heterogeneity in Sézary syndrome, a rare cutaneous T-cell lymphoma.

- Builds analysis pipelines from raw sequencing data through to biological interpretation
- Works independently to frame an open research question and carry it to a defensible result
- Collaborates with wet-lab and clinical teams to test computational predictions
- Project management: time management, prioritization across concurrent analyses and collaborators
- Open to postdoctoral, research, or industry roles in genomics, oncology, or single-cell biology

## EDUCATION

**PhD in Cancer Biology, Université de Bordeaux** ***Sep 2023 – Jun 2026***

*Research unit: INSERM U1312 | Bordeaux Research in Translational Oncology (BRIC)*

***Supervisors:*** *Prof. Audrey Gros (Director) and Dr. Slim Karkar (Co-Director)*

*Thesis defended 23 June 2026 | manuscript to be published online by end of August, 2026*

**Thesis:** "Deciphering Sézary syndrome tumoral heterogeneity through a bioinformatic approach" (see Research Experience for details)

**Masters in Bioinformatics, Uppsala University** ***Aug 2010 – Oct 2012***

*Sweden*

## RESEARCH EXPERIENCE

**PhD Researcher, Université de Bordeaux / INSERM U1312 (BRIC)** *2023 – 2026*

*Supervisors: Prof. Audrey Gros, Dr. Slim Karkar*

- Built a single-cell multi-omic pipeline (transcriptome, surface protein, TCR clonality) across seven patient-derived cell lines from three Sézary syndrome patients
- Combined whole-exome sequencing, BioNano optical mapping, and single-cell transcriptomics into one coherent model of tumor structure and behavior
- Resolved Th1-, Th2-, and Treg-like phenotypic states via pseudotime trajectory analysis against a healthy-donor CD4+ T-cell reference
- Identified transcription-factor-driven regulatory programs and a novel membrane biomarker; validated the biomarker's role in tumor cell survival via in vitro pharmacological inhibition with wet-lab colleagues

**Research Engineer, IRBI Insect Biology Research Institute** *2022 – 2023*

*Tours, France*

- Administered the group's HPC/Slurm cluster and long-read (PacBio, Nanopore) analysis workflows

**Research Engineer, INRAE** *2019 – 2021*

*Clermont-Ferrand, France*

- Benchmarked small RNA-seq tools and set the group's standard analysis pipeline
- Ran the group's Galaxy platform and delivered metatranscriptomic analyses on drug effects in animal models

**Research Assistant, Hong Kong University of Science and Technology (HKUST)** *2017 – 2018*

- Benchmarked exome and whole-genome sequencing pipelines for variant calling in cancer patient data

**Earlier research roles** *2009 – 2017*

- Marine prokaryotic genome analysis and web development | **Universidad Miguel Hernández de Elche**
- Behavioral genetics and epigenetics research | **Evolutionary Biology Centre, Sweden**
- Bacterial genomics, contributing to identification of a new Capnocytophaga species | **Karolinska Hospital, Sweden**
- Exome sequencing for rare disease variant discovery and involvement in Wetlab experiments | **COMSATS University**

## TEACHING

**Teaching Assistant (TP — Travaux Pratiques), Université de Bordeaux** *2023 – 2025*

- Delivered practical/laboratory teaching sessions (TP) each year of the PhD: 20 HETD equivalent hours in 2023–2024, 10 HETD equivalent hours in 2024–2025

**Instructor of Bioinformatics, Virtual University of Pakistan** *Jan 2017 – Jul 2017*

- Taught bioinformatics coursework and supervised laboratory exercises in molecular biology, microbiology, and cell biology

## PUBLICATIONS

***Under Review***

Boccelari L, Gravelle P, Tosolini M, **Abbasi N**, Le Morvan V, et al. Primary Cutaneous Diffuse Large B Cell Lymphoma, Leg Type, Induces Dynamic Transcriptional Reprogramming of Immunosuppressive Macrophages. *Cancer Immunology Research.* **Under review**

**Abbasi N**, et al. Deciphering Sézary syndrome tumoral heterogeneity through a bioinformatic approach. *To be submitted online, end of 2026.* ***Under review***

***In Preparation***

Cherfan C, Deslande M, Prochazkova-Carlotti M, **Abbasi N**, Sharanek A, Pham-Ledard A, Poglio S, Gros A, Merlio J-P, Bresson-Bepoldin L, Beylot-Barry M, Chevret E. Oxidative Phosphorylation: A Potential Achilles' Heel of Sézary Cells

***Published***

Ajmal M, Khan M, Neveling K, Tayyab A, Jaffar S, Sadeque A, Ayub H, **Abbasi N**, Riaz M, Micheal S, Gilissen C, Ali S, Azam M, Collin R, Cremers F, Qamar R. Exome sequencing identifies a novel and a recurrent BBS1 mutation in Pakistani families with Bardet-Biedl syndrome. *Molecular Vision.* 2013;19:644-653

Zangenah S, **Abbasi N**, Anders FA, Bergman P. Whole Genome Sequencing Reveals Novel Species of genus Capnocytophaga Isolated From Dog and Cat Bite Wounds in Humans. *Scientific Reports.* 2016

Zajitschek S, Herbert-Read JE, **Abbasi N**, Immler S. Paternal personality and social status influence offspring personality in offsprings. *BMC Evolutionary Biology.* 2017;17:157

## CONFERENCES & PRESENTATIONS

- “Single-cell dissection of Sézary syndrome heterogeneity” | Flash Talk & Poster, SBM Day conference, France (September 2024)
- “Tumor heterogeneity and regulatory programs in cutaneous lymphoma” | Poster, SBM Day conference, France (September 2025)
- Poster Presentation |Doctoral School Day, Université de Bordeaux, France (2025)
- gdec-bioinfo-workshop#2 | Tidyverse / R Markdown for Genomics, training workshop, France
- The Hong Kong Epigenomics Project | seminar series, Hong Kong
- The Mathematics of Deep Learning | seminar, Hong Kong

## LANGUAGES

**English:** Fluent | **French:** Professional working proficiency | **Urdu:** Native

## TECHNICAL & COMPUTATIONAL SKILLS

**Single-cell analysis:** Seurat, Scanpy, SingleCellExperiment, SCTransform, Harmony, DoubletFinder, Scrublet, Azimuth, SingleR, scType, Monocle3, Slingshot, PAGA, scVelo, Velocyto

**Spatial transcriptomics:** Seurat spatial, squidpy (self-directed, in progress)

**Genomics:** WES, WGS, RNAseq, Metatranscriptomics, Ampicon sequencing, BioNano optical mapping, variant calling, inferCNV, SCEVAN

**Regulatory & communication networks:** SCENIC, GENIE3, DoRothEA, LIANA, CellChat

**Statistics & differential expression:** DESeq2, edgeR, MAST, LIBRA (pseudobulk), Wilcoxon rank-sum test

**Visualization:** ggplot2, ComplexHeatmap, SCpubr, Nebulosa, dittoSeq

**Languages:** R, Python, Perl, C, JavaScript, PHP, SQL

**Infrastructure & tools:** HPC/Slurm administration, Snakemake, Conda, Docker, Singularity, Git, Galaxy administration, cloud computing

## REFERENCES

**References will be provided on request.**